

# Risk Register

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## Operational environment:

The bank is in a coastal area with low crime. It has 100 employees working in the office and 20 working remotely. The bank serves 2,000 personal customers and 200 business customers. It also works with local businesses and must follow strict financial regulations to protect customer information and bank funds.

## Risk Assessment

| Asset | Risk(s)                                                                                                                                                                                                                                                                                                                                          | Description                                                                          | Likelihood | Severity | Priority |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------|----------|----------|
| Funds | Business email compromise                                                                                                                                                                                                                                                                                                                        | <i>An employee may receive a phishing email and share confidential information.</i>  | 2          | 2        | 4        |
|       | Compromised user database                                                                                                                                                                                                                                                                                                                        | <i>Customer data is weakly protected, so malicious actors may gain access to it.</i> | 2          | 3        | 6        |
|       | Financial records leak                                                                                                                                                                                                                                                                                                                           | <i>A database server of backed up data is publicly accessible.</i>                   | 3          | 3        | 9        |
|       | Theft                                                                                                                                                                                                                                                                                                                                            | <i>Poor physical security could allow cash or valuable items to be stolen.</i>       | 1          | 3        | 3        |
|       | Supply chain disruption                                                                                                                                                                                                                                                                                                                          | <i>Storms or other natural events could delay important deliveries and services.</i> | 1          | 2        | 2        |
| Notes | <i>The bank works with many customers, employees, and outside companies, which creates different security risks. Human mistakes, cyberattacks, and natural disasters could affect bank operations and customer funds. The most important risks are protecting financial records and preventing unauthorized access to sensitive information.</i> |                                                                                      |            |          |          |

**Asset:** An asset is something valuable that the bank needs to protect. In this assessment, the main asset at risk is the bank's funds because they are important for daily operations and customer services.

**Risk(s):** A risk is anything that can impact confidentiality, integrity or availability of an asset. It is a possible event or threat that may damage the bank's systems, information, or funds. Examples of risks include phishing attacks, data breaches, theft, and supply chain disruptions.

**Description:** A vulnerability that might lead to a security incident.

**Likelihood:** Likelihood shows how possible it is for a risk to happen. The score is measured from 1 to 3:

- 1 = Low chance of happening
- 2 = Moderate chance of happening
- 3 = High chance of happening

**Severity:** Severity shows how much damage a risk could cause to the bank. The score is measured from 1 to 3:

- 1 = Low impact
- 2 = Moderate impact
- 3 = High impact

**Priority:** Priority shows which risks should be handled first. It is calculated by multiplying the likelihood score by the severity score:

$$\text{Likelihood} \times \text{Severity} = \text{Priority Score}$$

A higher priority score means the risk requires faster attention and more security resources.

## Sample risk matrix

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|            |              | Severity |               |                   |
|------------|--------------|----------|---------------|-------------------|
|            |              | Low<br>1 | Moderate<br>2 | Catastrophic<br>3 |
| Likelihood | Certain<br>3 | 3        | 6             | 9                 |
|            | Likely<br>2  | 2        | 4             | 6                 |
|            | Rare<br>1    | 1        | 2             | 3                 |

### Risk Matrix Explanation

The risk matrix helps the bank identify which risks need the most attention. The priority score is calculated by multiplying the likelihood of a risk happening by the severity of the impact.

- **Financial records leak ( $3 \times 3 = 9$ ):** This is the highest priority because it is very likely and could cause serious damage to the bank and its customers.
- **Compromised user database ( $2 \times 3 = 6$ ):** This is a high-risk issue because attackers gaining access to customer data could cause financial and reputational damage.
- **Business email compromise ( $2 \times 2 = 4$ ):** This is a moderate risk because phishing attacks can happen, but the impact may be lower compared to a major data breach.
- **Theft ( $1 \times 3 = 3$ ):** This risk has a lower priority because theft is less likely in a low-crime area, although the impact could still be serious.
- **Supply chain disruption ( $1 \times 2 = 2$ ):** This has the lowest priority because it is less likely and would have a moderate effect on operations